

Deterministic pose-spline dynamics estimator

Shared parameter update: nominal -> estimated

mass [kg]	2.351597591 -> 2.761922494	delta=+0.410325
ratio=1.1744877		
CoG x [m]	0 -> -0.001223043103	delta=-0.00122304 ratio=-
CoG y [m]	0 -> 0.002309951388	delta=+0.00230995 ratio=-
CoG z [m]	0 -> -0.02762725296	delta=-0.0276273 ratio=-
rotor effectiveness 1	1 -> 0.9833530397	delta=-0.016647 ratio=0.98335304
rotor effectiveness 2	1 -> 1.057453341	delta=+0.0574533 ratio=1.0574533
rotor effectiveness 3	1 -> 0.9746359336	delta=-0.0253641 ratio=0.97463593
rotor effectiveness 4	1 -> 0.9867040316	delta=-0.013296 ratio=0.98670403
command lag [s]	0.01 -> 0.04275679864	delta=+0.0327568 ratio=-

Inertia matrix [kg m²]

nominal:
[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]
estimated:
[0.06909224285, -0.002242554403, 1.760606701e-05]
[-0.002242554403, 0.16834866, -0.0001473277053]
[1.760606701e-05, -0.0001473277053, 0.2364764859]

Principal moments: nominal -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.06904160022	delta=+0.00408896
ratio=1.0629529		
principal moment 2 [kg m ²]	0.06500006701 -> 0.168398981	delta=+0.103399
ratio=2.5907509		
principal moment 3 [kg m ²]	0.1289921163 -> 0.2364768075	delta=+0.107485
ratio=1.8332656		

Optimization

joint dynamics loss: 0.251462884605
soft-prior cost: 0.0639975885038
selected strict-ZOH lag [s]: 0.042756798636

Bag failure_1_single

reconstruction position RMSE [m]: 0.000511441948808
reconstruction position component RMSE [m]: [0.000264 0.000287 0.000331]
reconstruction orientation RMSE [deg]: 0.491033445648
gyro consistency RMSE xyz [rad/s]: [0.050326 0.085763 0.038402]
specific-force consistency RMSE xyz [m/s²]: [0.941624 0.163374 0.348233]
inferred external force RMS xyz [N]: [0.628769 0.310638 1.215816]
inferred external torque RMS xyz [N m]: [0.059489 0.476723 0.144709]
bag dynamics loss: 0.251462884605